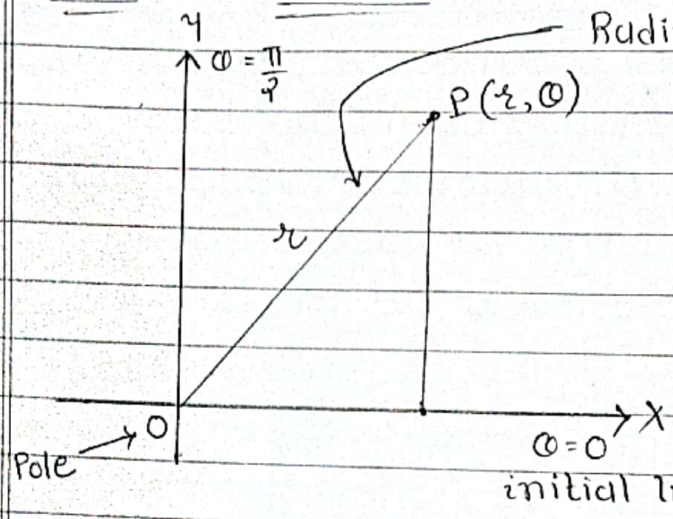
Points of Intersection:

Determine the points where the curve meets the initial line $\theta = 0, \pi/2$ & π.

* Polar co-ordinates:



Radius vector OR Polar

(r, θ) - Polar co-ordinate

* Procedure of tracing polar curve

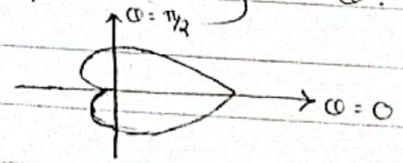
Consider the curve equation $f(r, \theta) = 0$

(1) Symmetry:

The curve is symmetric about

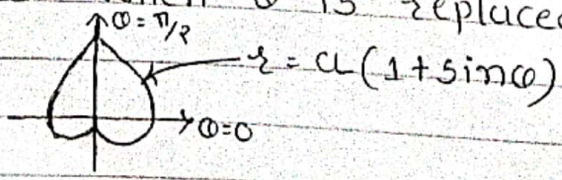
a) the initial line $\theta = 0$ as the equation remains unchanged when θ is replaced by $-\theta$.

For Ex: $r = a(1 + \cos\theta)$



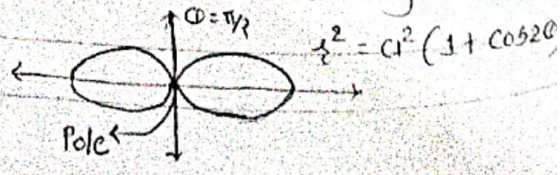
b) the line $\theta = \pi/2$ as the equation remains unchanged when θ is replaced by $(\pi - \theta)$.

For Ex:



(c) the pole as the equation remains unchanged when r is replaced by $-r$.

θ by $\pi + \theta$



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d) about the line $\theta = \pi/4$, replacing θ by $\frac{\pi}{2} - \theta$

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2) Pole :

a) curve passes through the pole if $r = 0$, for some value θ (say $\theta = \alpha$)

b) Tangent at the pole is given by $\theta = \alpha$

For Ex: Curve $r = a(1+\cos\theta)$

$r = 0 \Rightarrow a(1+\cos\theta) = 0$

$\Rightarrow \cos\theta = -1$

$\Rightarrow \theta = \pi$

∴ curve passes through the pole.
and $\theta = \pi$ (initial line) is tangent at the pole.

3) Table :

Prepare a table for values of θ and corresponding values of r is useful in plotting a polar curve.

(4) Region :

(a) Find max. value of r, so we say that the entire curve lies within the circle.

For Ex.: The curve (cardioid) $r = a(1+\cos\theta)$

Here max. value of $r = 2a$

So, the entire curve lies within a circle $r = 2a$.

(b) Find the value of θ for which $r^2 < 0$, so that the curve does not exist for these values of θ.
(If r occurs in even powers)

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4) Direction of tangent:

$r = a(1+\cos\theta)$

$\frac{dr}{d\theta} = -a\sin\theta$

Now, $\tan\phi = r\,d\theta/dr$

$\tan\phi = \frac{a(1+\cos\theta)}{-a\sin\theta}$

$= \frac{-2\sin^2\theta/2}{-2\sin\theta/2\cos\theta/2}$

$= \tan\theta/2$

$\phi = \theta/2$

Ex: 1 Trace the cardioid $r = a(1+\cos\theta)$

Sol^n:

∴ At pt. $A(a, \pi/2)$, $\phi = \pi/4$

∴ tangent makes an angle $\pi/4$ with $\theta = \pi/2$ line

(1) Symmetry:

If we replace θ by $-\theta$ then equation remains same.

∴ The curve is symmetric about the initial line $\theta = 0$.

(2) Pole:

$r = 0 \Rightarrow a(1+\cos\theta) = 0 \Rightarrow \cos\theta = -1 \Rightarrow \theta = \pi$

∴ curve passes through the pole.

→ $\theta = \pi$ is the tangent at the pole.

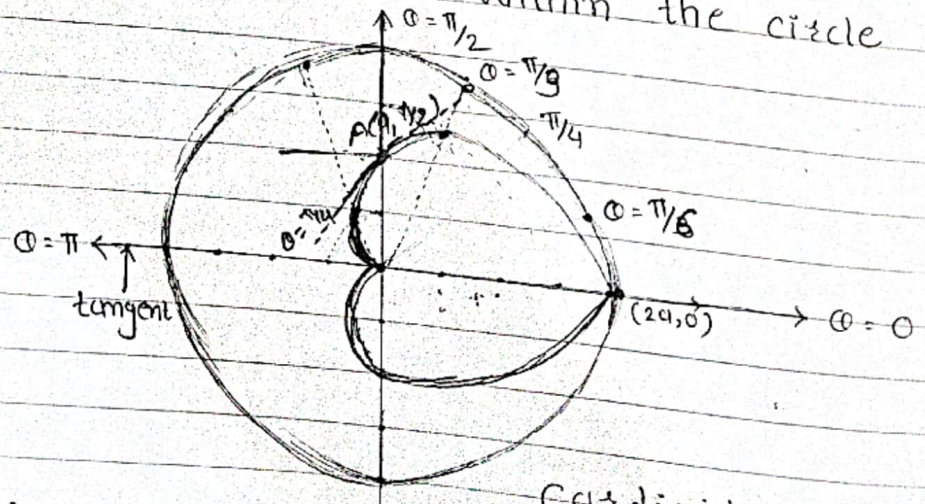
(3) Table:

θ	0	$\pi/6$	$\pi/4$	$\pi/3$	$\pi/2$	$2\pi/3$	π
r	$2a$	$a(1+\sqrt{3}/2)$	$a(1+1/\sqrt{2})$	$3a/2$	a	$a/2$	0
		1.465	1.709				
		2.865	2.709				

(4) Region:

From the table, the max. value of $r = 2a$

∴ The entire curve lies within the circle $r = 2a$.



Cardioid

3) Pts. of intersection: Putting $\theta = 0, \pi/2, \pi$,

$r = 2a, a, 0$.

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$2\pi - \pi/6$

$11\pi/6$

$\frac{\sin 11\pi/6}{\sin(2\pi - \pi/6)} \quad \frac{-\sin \pi/6}{1 - 1/2}$

Ex: 2 Trace the curve $r = a(1 + \sin\theta)$

Sol^{n}:

(1) Symmetry:

If we replace θ by $\pi - \theta$, the equation remains same.

$\therefore$ curve is symmetric about the line $\theta = \pi/2$.

(2) Pole:

$$r = 0 \Rightarrow a(1 + \sin\theta) = 0$$
$$\Rightarrow \sin\theta = -1$$
$$\Rightarrow \theta = 3\pi/2$$

$\therefore$ curve passes through the pole

$\rightarrow \theta = 3\pi/2$ is tangent at the pole.

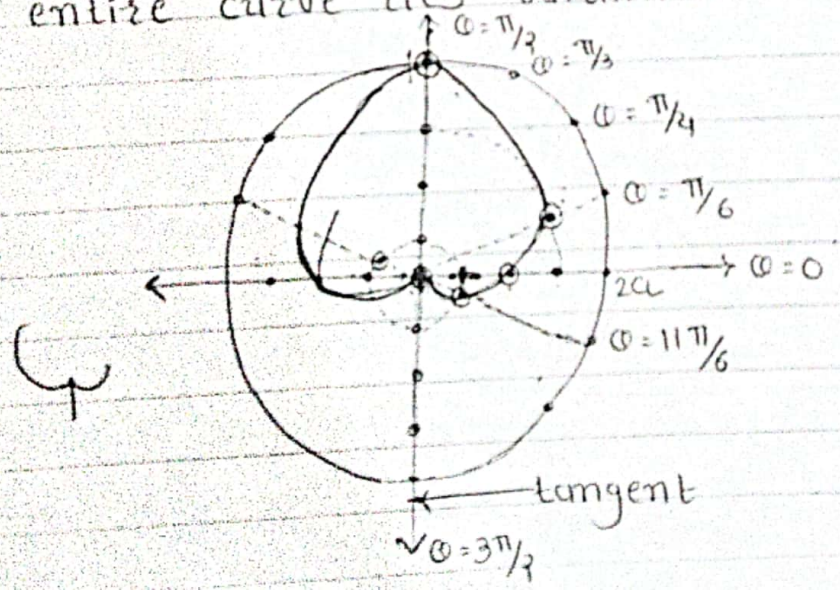
(3) Table:

θ	0	$\pi/6$	$\pi/4$	$\pi/3$	$\pi/2$	$11\pi/6$	$3\pi/2$
r	a	$3a/2$	$a(1+1/\sqrt{2})$	$a(1+\sqrt{3}/2)$	$2a$	$a/2$	0

(4) Region:

From the table, max. value of $r = 2a$

$\therefore$ The entire curve lies within the circle $r = 2a$.



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$r = a\sin n\theta$ } if n is odd, n petal
$r = a\cos n\theta$ } if n is even, $2n$ petal

[3] Trace the curve $r = a\sin 3\theta$ (Three petal rose)

Sol^n:

(1) Symmetry:

If we replace θ by $\pi - \theta$, then equation remains same.

$\therefore$ Curve is symmetric about the line $\theta = \pi/2$.

(2) Pole:

$$r = 0 \Rightarrow a\sin 3\theta = 0$$
$$\Rightarrow 3\theta = 0, \pi, 2\pi, 3\pi, 4\pi, 5\pi, \ldots$$
$$\Rightarrow \theta = 0, \frac{\pi}{3}, \frac{2\pi}{3}, \pi, \frac{4\pi}{3}, \frac{5\pi}{3}, \ldots$$

$\therefore$ Curve passes through the origin pole.
$\rightarrow \theta = 0, \frac{\pi}{3}, \frac{2\pi}{3}, \ldots$ are tangent at the pole.

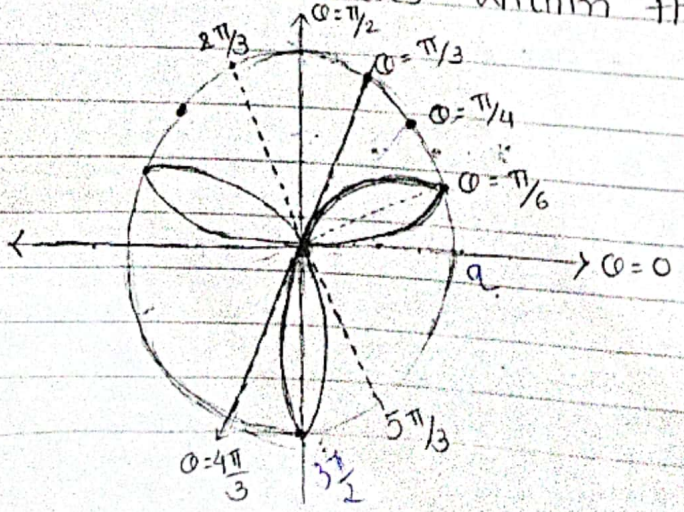
(3) Table:

θ	0	$\pi/6$	$\pi/4$	$\pi/3$	$\pi/2$	$11\pi/6$	$5\pi/3$	$3\pi/2$
r	0	a	$a/\sqrt{2}$ (0.70)	0	$-a$	$-a$	0	a

(4) Region:

From the table, max. value of $r = a$

$\therefore$ The entire curve lies within the circle $r = a$



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[4] Trace the curve $r = a\cos 2\theta$

(1) Symmetry:

If we replace θ by $-\theta$ and $\pi - \theta$, then equation remains same.

∴ Curve is symmetric about the initial line $\theta = 0$ and the line $\theta = \pi/2$.

(2) Pole:

$$r = 0 \Rightarrow a\cos 2\theta = 0$$
$$\Rightarrow 2\theta = \pi/2, 3\pi/2, 5\pi/2, 7\pi/2, \ldots$$
$$\Rightarrow \theta = \pi/4, 3\pi/4, 5\pi/4, 7\pi/4, \ldots$$

∴ curve passes through the origin.

and $\theta = \pi/4, 3\pi/4, 5\pi/4, \ldots$ are tangent at the pole.

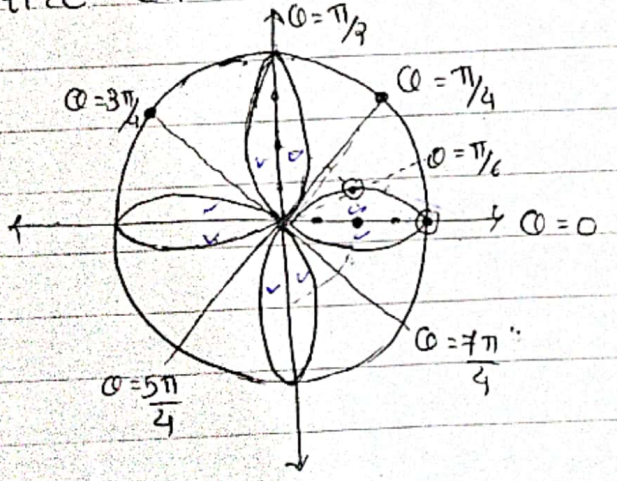
(3) Table:

θ	0	$\pi/6$	$\pi/4$	$\pi/3$	$\pi/2$	$3\pi/4$
r	a	$a/2$	0	$-a/\sqrt{2}$	$-a$	0

(4) Region:

From the table, max. value of $r = a$

∴ The entire curve lies within $r = a$



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[5] Trace Lemniscate of Bernoulli $r^2 = a^2 \cos 2\theta$.

Sol^n:

(1) Symmetry:

Curve is symmetric about the initial line $\theta = 0$, the line $\theta = \pi/2$ and the pole.

(2) Pole:

$$r = 0 \Rightarrow r^2 = 0$$
$$\Rightarrow a^2 \cos 2\theta = 0$$
$$\Rightarrow 2\theta = \pi/2, 3\pi/2, \ldots$$
$$\Rightarrow \theta = \pi/4, 3\pi/4, \ldots$$

∴ curve passes through the pole and $\theta = \pi/4, 3\pi/4, \ldots$ are tangent at the pole.

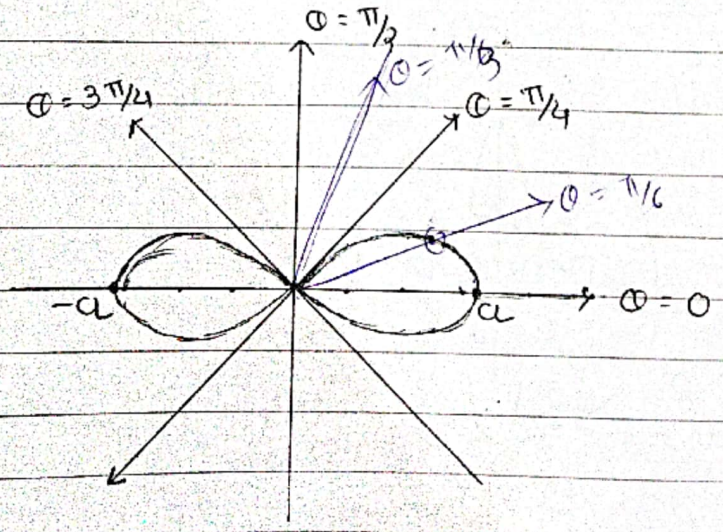
(3) Table:

θ	0	$\pi/6$	$\pi/4$	$\pi/3$	$\pi/2$	$3\pi/4$	$5\pi/6$	π
r	a	$a/\sqrt{2}$	0	x	x	0	x	$-a$
		0.7a						

(4) Region:

From the table, max. value of $r = a$.

∴ The entire curve lies within $r = a$.



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[6] Trace the curve $r = a\cos 3\theta$

Sol^{n}:

(1) Symmetry:

Initial line $\theta = 0$

(2) Pole: Take $r = 0$

$\Rightarrow a\cos 3\theta = 0$

$\Rightarrow 3\theta = \pi/2, 3\pi/2, 5\pi/2, 7\pi/2, \ldots$

$\Rightarrow \theta = \pi/6, \pi/2, 5\pi/6, 7\pi/6, \ldots$

$\therefore$ curve passes through the pole.

and $\theta = \pi/6, \pi/2, \ldots$ are tangent at the pole.

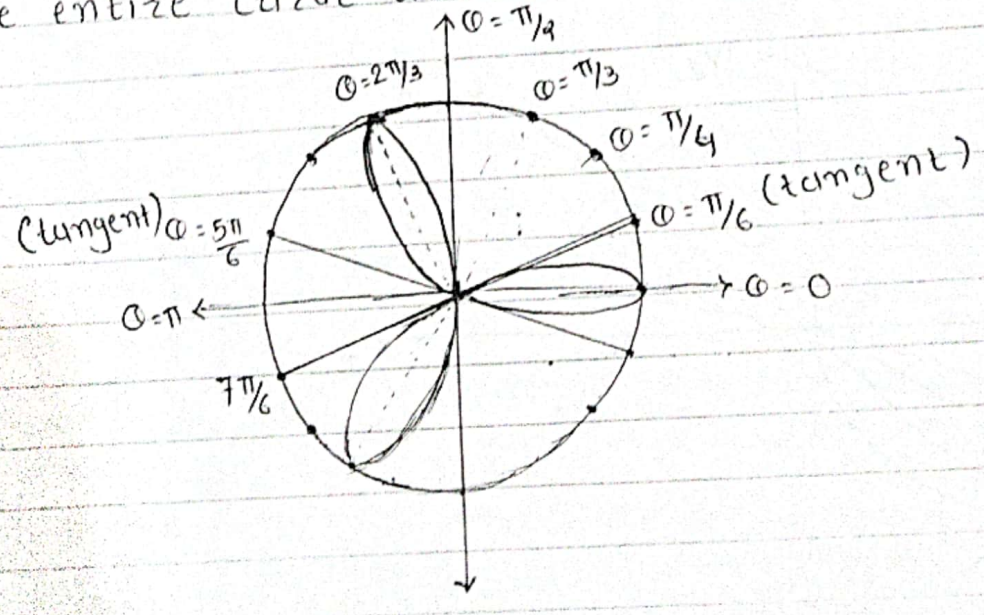
(3) Table:

θ	0	$\pi/6$	$\pi/4$	$\pi/3$	$\pi/2$	$2\pi/3$	$5\pi/6$	π
r	a	0	$-a/\sqrt{2}$	$-a$	0	a	0	$-a$

(4) Region:

From the table, max. value of $r = a$

$\therefore$ The entire curve lies within the circle $r = a$.



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Trace the lemniscate $r^2 = a^2 \sin 2\theta$

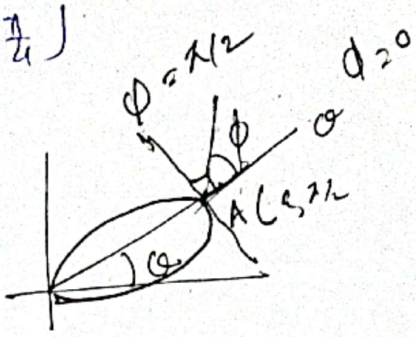
1) symmetry - about pole, also $\theta = \frac{\pi}{4}$

2) Pole: $r = 0$ at $\theta = 0, \pi/2$

$\therefore \theta = 0$ & $\theta = \frac{\pi}{2}$ are tangents at the pole.

3) Point of intersection:

at $\theta = \frac{\pi}{4}$, pt. of intersection is $A(a, \frac{\pi}{4})$



4) Direction of tangent; $r^2 = a^2 \sin 2\theta$

$$2r \frac{dr}{d\theta} = a^2 2\cos 2\theta$$

$$\frac{d\theta}{dr} = \frac{a^2 \cos 2\theta}{r}$$

Now, $\tan \phi = r \frac{d\theta}{dr} = \frac{r^2 \cdot a^2 \cos 2\theta}{r \cdot \cos 2\theta} = \frac{a^2 \sin 2\theta}{a^2 \cos 2\theta} = \tan 2\theta$

$\therefore \phi = 2\theta$

At $A(a, \frac{\pi}{4})$, $\phi = \frac{\pi}{2}$

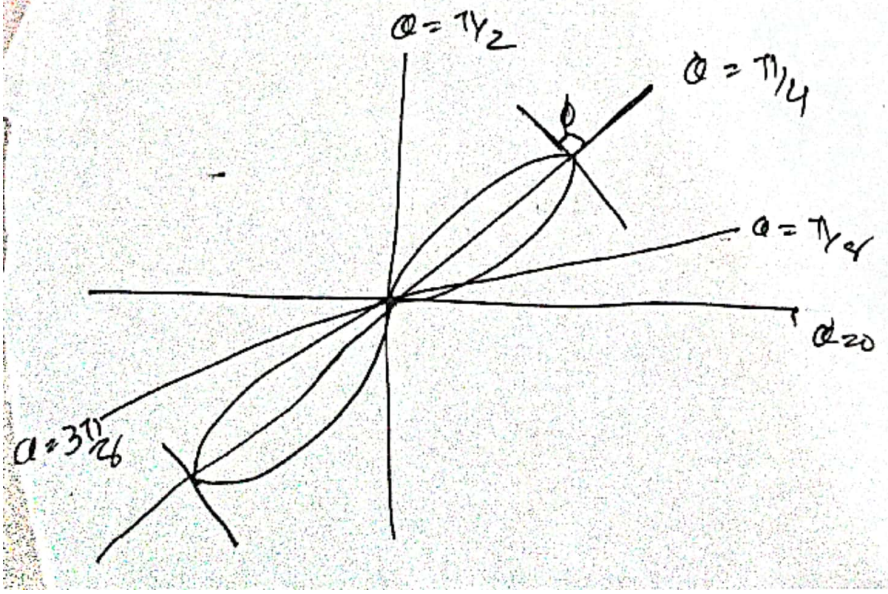
$\therefore$ The tangent is perpendicular to the line $\theta = \frac{\pi}{4}$.

Region:

θ	0	$\frac{\pi}{8}$	$\frac{\pi}{4}$	$\frac{3\pi}{8}$	$\frac{\pi}{2}$
r	0	$\frac{a}{(2)^{1/4}}$	a	$\frac{a}{2^{1/4}}$	0

Max. value of $\sin 2\theta$ is 1.

Max. value of r is a



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